

DEVICES AND CIRCUITS FOR ARTIFICIAL INTELLIGENCE
Project
Sports image classification

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Source code in .txt and .pdf:



Table of Content

- 1.0 Problem
- 2.0 Objective
- 3.0 Technique Used
- 4.0 Input, Process and Output of the system
- 6.0 Conclusion
- 7.0 References

1.0 Problem

Hundreds of sports exist, for some games players should participate in single contestants, through for some with hundreds of simultaneous participants in teams playing as individuals. As there are so many types of sports worldwide, it's quite difficult for people to know the names for all types of sports. Problems will arise when there are those who want to venture into the field of sports because it looks very attractive after watching it on television but do not know the name or category of the sport. Also, because there are several sports that look similar but are different, make it quite difficult for people to differentiate them. As computer science students, we may try to help, to solve this problem by creating a sports classification system. This system helps people to identify the criteria of some unknown sports so that all people can know the names of sports they like.

2.0 Objective

To identify sports in each category for classification.

3.0 Technique

the best neural network to complete image classification is the Convoluted Neural Network (CNN). reasons why I chose Neural Networks for project is because Convolutional Neural Networks use filters to create feature maps and include the presence of detected features in the input. Convolutional Neural Networks are automatic feature extractors from images without losing much of the spatial interaction between pixels. Lack of learning will reduce the accuracy of the results. picture filters are defined as neural networks that use deep learning and are designed to reduce processing requirements. And processing designed specifically to process pixel data. CNN consists of three types of layers namely input, output, and hidden layers. The hidden layers

on the Convolutional Neural Network include the unification layer, the convolution layer, and the Rectified Linear Units layer. CNN define the network architecture in a more sensible way and developer can arrange layers randomly in the state of at least one layer of each type. the central point or main building block of the CNN that controls the movement of the convulsive neural network is the convolution layer. Each hidden layer consists of a set of neurons, where each neuron is fully connected to all neurons in the previous layer. On the convolutional layer, it is responsible for detecting features and is done by applying a filter known as the 'kernel' over the image to extract those features. The type of convolution used is a 2-Dimension(2D) convolution layer. Filters or kernels in the convolutional 2D layer are over 2D input data and performing element multiplication. A filter or kernel in a conventional 2D layer is more than 2D input data whose input data size must be larger than the filter size and perform element multiplication such as 1x1, 2x2, and so on. The kernel will perform the same operation for each location launched, converting a 2D feature matrix into a different 2D feature matrix. So, we can consider a 5x5 image in a box whose pixel values are only 0 and 1. In addition, it can achieve sophisticated and impressive results. The next type of layer that is included in the Convolutional Neural Network is the Rectified Linear Units (Relu) layer. Rectified Linear Units is a linear function of a piece that will produce a direct input which if its input is positive, otherwise it will produce zero if its pixel values are negative. The purpose of using the substitution function is to correct the accuracy of our image. The reason we want to do this is that the picture is naturally nonlinear, and the convolution is a linear operation. Most images naturally consist of non-linear features such as borders, colors, the transition between pixels, and so on. The purpose of applying Relus is to increase the non-linearity in the images. The nonlinear activation function is the most preferred function to perform because it allows nodes to learn more complex structures in the data. Both functions can only process values between -1 to 1 while Relus has a larger range of pixel values and will have better predictive performance. The last type of layer in the Convolutional Neural Network is the pooling layer. And known as down sampling which reduces the size of the matrix. There is a difference between the convolution layer and the pooling layer. The convolution layer encapsulates the presence of features in the input image where the input image needs to be multiplied by the kernel to produce one value. The pooling layer provides an approach to derive the sampling feature map by summarizing the presence of features in the feature. By performing pooling operations, a similar pattern is produced but in a smaller size. There are three functions used in consolidation operations. First, averages pooling to calculate the average value or pixel for each patch on the feature map. The second is the minimum pooling to calculate the minimum value or pixel for each feature map patch. And the last one is the maximum pooling to calculate the maximum value or pixel for each feature map patch. The pixels are taken as a representation of that part of the picture. Among these three types of merging methods, I've have chosen maximum pooling for the project because maximum pooling is the most used because its high point is the brightest(sharpest) features of the picture in order to model the input representation and reduce its dimensions. And this method is more suitable for sport classification based on available data. If we want to choose a picture that has a dark background, then it is better to choose the maximum pooling because it chooses the brightest pixels.

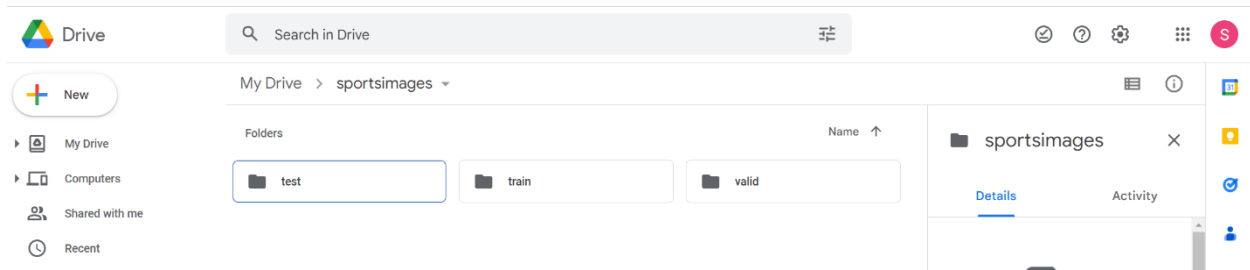
For the project I've used the ResNet-50 model. The Deep residual network, ResNet-50 model is a **convolutional neural network (CNN)** that is 50 layers deep. It has 48 Convolution layers along with 1 Max Pool and 1 Average Pool layer. Finally, the network has an Average Pooling layer followed by a fully connected layer having 1000 neurons (ImageNet class output). with a 3-layer bottleneck block, forming the Resnet 50 architecture. Deep residual nets make use of residual blocks to improve the accuracy of the models. A residual block is a stack of layers set in such a way that the output of a layer is taken and added to another layer deeper in the block. It has been observed that training the neural networks becomes more difficult with the increase in the number of added layers. ResNet was created with the aim of tackling this exact problem. With Resnet, it becomes possible to surpass the difficulties of training very deep neural networks. the network was inspired by VGG neural networks (VGG-16, VGG-19), with the convolutional networks having 3x3 filters. ResNets have fewer filters and lower complexity. the architecture of ResNet50. The architecture of ResNet50 has 4 stages. The network can take the input image having height, width as multiples of 32 and 3 as channel width. For the sake of explanation, we will consider the input size as 256 x 256 x 3. The size of kernels used to perform the convolution operation in all 3 layers of the block of stage 1 are 64, 64 and 128 respectively., the size of input will be reduced to half in terms of height and width, but the channel width will be doubled. As we progress from one stage to another, the channel width is doubled, and the size of the input is reduced to half. For each residual function F, 3 layers are stacked one over the other. The three layers are 1x1, 3x3, 1x1 convolutions. The 1x1 convolution layers are responsible for reducing and then restoring the dimensions. The 3x3 layer is left as a bottleneck with smaller input/output dimensions. ResNet50V2 model was initialized with pre-trained ImageNet weights. Only the Dense layers were fine-tuned.

- Two hidden layers of 256 and 128 neurons respectively were added with relu activations.
- Dropout Layers with p=0.1 were added to prevent overfitting.
- Number of epochs was 50 with Early stopping with patience parameter as 2 epochs.
- Batch size of 32 was used for training.
- Categorical Cross-Entropy was used as the loss function.

4.0 input process and output

Input

For the project I've used the dataset of 100 sport classes, a Collection of sports images covering 100 different sports. Images are 224,224,3 jpg format. Below figure shows how Data is separated into train, test and valid directories. And the second image shows the datasets of the project. Then I imported the required libraries. Then to apply augmentations I loaded the image datasets.



```
Found 13582 files belonging to 100 classes.  
Found 500 files belonging to 100 classes.  
Found 500 files belonging to 100 classes.
```

Process

I used Google colab to run the source code. Since the training dataset is ready let's create a simple CNN Model to train on the image datasets. After creating the model, I defined Callbacks, A callback is an object that can perform actions at various stages of training (e.g., at the start or end of an epoch, before or after a single batch) Then I compile and trained the model. And I set the number of epochs to 15 that means there are 15 passes through entire training data sets. Below shows how I set up the epoch to 15 to make sure the output accuracy is high.

```
resnet = ResNet50V2(  
    input_shape=(256, 256, 3), include_top=False, weights="imagenet", classes=100  
)  
resnet.trainable = False  
model = Sequential()  
model.add(resnet)  
model.add(Flatten())  
model.add(Dropout(0.1))  
model.add(Dense(256, activation="relu"))  
model.add(Dropout(0.1))  
model.add(Dense(128, activation="relu"))  
model.add(Dropout(0.1))  
model.add(Dense(100, activation="softmax"))  
  
model.compile(  
    optimizer="adam", loss=CategoricalCrossentropy(), metrics=["accuracy"]
```

```

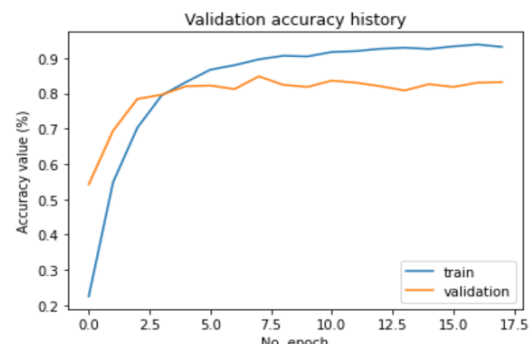
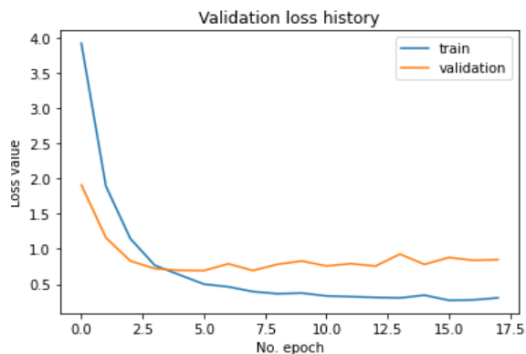
callback = EarlyStopping(monitor="loss", patience=2)
history = model.fit(
    train_generator,
    epochs=15,
    batch_size=32,
    callbacks=[callback],
    validation_data=validation_generator,
    \

```

Output

Due to the usage of two data sets the experiment results in two types of accuracy which are training accuracy and test accuracy. Training accuracy measures the effectiveness of the model to learn, and testing accuracy evaluate the performance of the model. the model learns more about noise or any distraction characteristics in the image that are not wanted by the receiver. Plotting training and validation accuracy, loss.

To achieve these results. I changed our batch size from 256 to a batch size of 32. We used the optimizer, Adam. We run our algorithm for 15 epochs with categorical cross-entropy loss, and we achieved an accuracy of 84.91% and a loss of 1.9092. Time taken for the training process was 1m 48s. Since we experienced overfitting in the training set, we added dropout layers for regularization.



Performance metrics that we use to assess the quality of the model are, Precision: Percentage of correct positive predictions relative to total positive predictions. Recall: Percentage of correct positive predictions relative to total actual positives. And F1 Score: A weighted harmonic mean of precision and recall. The closer to 1, the better the model. I got these metrics by computing the classification report. F1 Score is calculated by: $2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

Using these three metrics, we can understand how well a given classification model is able to predict the outcomes. Support in the classification report depicts the number of actual occurrences of the class in the dataset.

	precision	recall	f1-score	support					
air hockey	0.83	1.00	0.91	5					
ampute football	0.83	1.00	0.91	5	frisbee	0.75	0.60	0.67	5
archery	1.00	0.80	0.89	5	gaga	1.00	0.20	0.33	5
arm wrestling	1.00	1.00	1.00	5	giant slalom	0.83	1.00	0.91	5
axe throwing	1.00	1.00	1.00	5	golf	0.50	0.60	0.55	5
balance beam	0.83	1.00	0.91	5	hammer throw	1.00	1.00	1.00	5
barrell racing	1.00	0.80	0.89	5	hang gliding	1.00	1.00	1.00	5
baseball	1.00	0.40	0.57	5	harness racing	1.00	1.00	1.00	5
basketball	0.67	0.40	0.50	5	high jump	0.71	1.00	0.83	5
baton twirling	1.00	0.80	0.89	5	hockey	0.71	1.00	0.83	5
bike polo	1.00	0.80	0.89	5	horse jumping	1.00	1.00	1.00	5
billiards	1.00	1.00	1.00	5	horse racing	1.00	1.00	1.00	5
bmX	0.80	0.80	0.80	5	horseshoe pitching	1.00	0.60	0.75	5
bobsled	1.00	0.60	0.75	5	hurdlles	1.00	1.00	1.00	5
bowling	1.00	0.60	0.75	5	hydroplane racing	1.00	0.40	0.57	5
boxing	0.71	1.00	0.83	5	ice climbing	0.80	0.80	0.80	5
bull riding	1.00	1.00	1.00	5	ice yachting	1.00	1.00	1.00	5
bungee jumping	0.56	1.00	0.71	5	jai alai	0.67	0.80	0.73	5
canoe slamon	0.71	1.00	0.83	5	javelin	1.00	1.00	1.00	5
cheerleading	1.00	0.60	0.75	5	jousting	0.83	1.00	0.91	5
chuckwagon racing	1.00	0.80	0.89	5	judo	0.83	1.00	0.91	5
cricket	0.71	1.00	0.83	5	lacrosse	0.60	0.60	0.60	5
croquet	0.83	1.00	0.91	5	log rolling	0.83	1.00	0.91	5
curling	1.00	1.00	1.00	5	luge	0.83	1.00	0.91	5
disc golf	1.00	0.80	0.89	5	motorcycle racing	0.71	1.00	0.83	5
fencing	1.00	1.00	1.00	5	mushing	1.00	1.00	1.00	5
field hockey	0.80	0.80	0.80	5	nascar racing	1.00	1.00	1.00	5
figure skating men	1.00	1.00	1.00	5	olympic wrestling	0.62	1.00	0.77	5
figure skating pairs	0.75	0.60	0.67	5	parallel bar	0.75	0.60	0.67	5
figure skating women	0.80	0.80	0.80	5	pole climbing	1.00	0.80	0.89	5
fly fishing	1.00	1.00	1.00	5	pole dancing	0.80	0.80	0.80	5
football	0.57	0.80	0.67	5	pole vault	0.50	0.40	0.44	5
formula 1 racing	1.00	1.00	1.00	5	polo	1.00	1.00	1.00	5
					pommel horse	0.71	1.00	0.83	5
					rings	1.00	0.80	0.89	5
					rock climbing	0.83	1.00	0.91	5
					roller derby	1.00	1.00	1.00	5
					rollerblade racing	0.67	0.80	0.73	5
					rowing	0.75	0.60	0.67	5
rugby	0.80	0.80	0.80	5					
sailboat racing	0.71	1.00	0.83	5					
shot put	0.67	0.80	0.73	5					
shuffleboard	1.00	0.80	0.89	5					
sidecar racing	1.00	0.80	0.89	5					
ski jumping	0.62	1.00	0.77	5					
sky surfing	0.50	0.40	0.44	5					
skydiving	0.71	1.00	0.83	5					
snow boarding	0.33	0.20	0.25	5					
snowmobile racing	1.00	0.80	0.89	5					
speed skating	1.00	0.80	0.89	5					
steer wrestling	0.83	1.00	0.91	5					
sumo wrestling	0.80	0.80	0.80	5					
surfing	1.00	0.60	0.75	5					
swimming	1.00	0.80	0.89	5					
table tennis	0.71	1.00	0.83	5					
tennis	1.00	0.80	0.89	5					
track bicycle	1.00	0.80	0.89	5					
trapeze	1.00	0.60	0.75	5					
tug of war	0.57	0.80	0.67	5					
ultimate	1.00	0.40	0.57	5					
uneven bars	1.00	0.80	0.89	5					
volleyball	1.00	0.80	0.89	5					
water cycling	1.00	1.00	1.00	5					
water polo	0.83	1.00	0.91	5					
weightlifting	1.00	1.00	1.00	5					
wheelchair basketball	0.83	1.00	0.91	5					
wheelchair racing	0.83	1.00	0.91	5					
wingsuit flying	1.00	1.00	1.00	5					
accuracy			0.84	500					
macro avg	0.86	0.84	0.84	500					
weighted avg	0.86	0.84	0.84	500					

6.0 Conclusion

Overall, accuracy is one of the important aspects to measure the successfulness of the image classification but it does not always give the full picture. The accuracy in fact is less likely not suitable to be used for the experiments with imbalance sample of data.

In conclusion, this project discussed how CNN was able to classify images using Keras API and tensorflow. From the findings, CNN is exactly the best option to classify the image due to its intelligence and flexibility.

7.0 References

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